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Received: 16 May 2020 | Revised: 22 December 2020 | Accepted: 26 December 2020

DOI: 10.1111/jocs.15304

ORIGINAL ARTICLE

JOURNAL OF CARDIAC SURGERY WILEY

Major adverse cardiac events and functional capacity in patients at intermediate risk undergoing transcatheter versus surgical aortic valve replacement for aortic stenosis with bicuspid valves

Hui-Yu Tsai BS | Yu-Shan Lin MS | I-Chen Wu BS | Li-Ying Kuo MS | Bo-Yan Chen BS | Shiang-Lin Shen BS | Wei-Ning Hsu BS | Hsin-Yi Huang MS

Heart Center, Cheng-Hsin General Hospital, Peitou, Taipei, Taiwan

Correspondence
Hsin-Yi Huang, MS, Heart Center, Cheng-Hsin General Hospital, No. 45 Cheng-Hsin St., Peitou, Taipei, Taiwan.
Email: 856038@gmail.com

Abstract

Background: Transcatheter aortic valve replacement (TAVR) is not always the optimal option for aortic valve stenosis (AS) patients with bicuspid aortic valves (BcAVs) and many studies exclude this group of patients. The aim of our study was to compare the rate of a major adverse cardiovascular event (MACE) and functional capacity in AS patients with BcAV after surgical aortic valve replacement (SAVR) and TAVR.

Methods: This study included 130 patients who underwent SAVR or TAVR from July 2013 to August 2018 at the Cheng Hsin General Hospital. The main outcome was MACE. Events recorded included noncardiovascular (CV) mortality, CV mortality, recurrent nonfatal stroke, recurrent nonfatal myocardial infarction (MI), and important events. The secondary outcome was functional recovery, which was defined according to the metabolic equivalent (MET) 6 months after the aortic procedure.

Results: The mean age of patients was 56.8 ± 26.9 years and the mean Society of Thoracic Surgeons score was 3.29 ± 4.69. Logistic regression analyses indicated that SAVR was a significant predictor of functional recovery. Patients who underwent SAVR had a higher rate of functional recovery (>3 METs; 87.8%, $p = .000$) and had a significantly higher odds ratio (3.56; 95% confidence interval, 1.19–10.63, $p = .023$). The Kaplan–Meier survival analysis showed that the MACE rate was not associated with the aortic procedure.

Conclusions: Our analysis showed that SAVR is a significant predictor of better functional recovery and TAVR is associated with a lower level of functional capacity. In summary, TAVR is an acceptable option for AS patients with BcAV, and for a better prognosis, an early intervention aimed at improving functional capacity is highly recommended for this group of patients.

KEYWORDS

aortic stenosis, bicuspid valves, cardiac events, functional capacity, transcatheter aortic valve replacement

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1 | INTRODUCTION

Nowadays, transcatheter aortic valve replacement (TAVR) plays an important role in the management of symptomatic severe aortic valve stenosis (AS) in patients who are considered not suitable for conventional surgery due to the high risk. Because of the risk of severe comorbidities and complications, the heart team needs to perform risk stratification using validated risk scores, such as the European System for Cardiac Operative Risk Evaluation (EuroSCORE) and the Society of Thoracic Surgeons (STS) score,[1] and make a decision before performing TAVR.

Although previous research and guidelines have established TAVR as the standard treatment in high surgical risk patients,[2] TAVR is not always the optimal option for patients with bicuspid aortic valves (BcAVs) because of the specific anatomic characteristics of BcAV.[3] A large research study, the placement of aortic transcatheter valves 3 trial, compared outcomes of surgical aortic valve replacement (SAVR) and TAVR in patients with severe AS. This large trial excluded AS patients with BcAV because the morphological characteristics of BcAV may be associated with a higher rate of complications after surgery, even in patients who are considered low risk.[4]

However, because BcAV is one of the most prevalent valvular abnormalities in adults, the prognosis after surgery is important in this special group. Some studies have shown that TAVR is feasible in selected AS patients with BcAV, but unfortunately, studies have lacked a control group of patients with BcAV who received SAVR to facilitate comparison.[5–8] Furthermore, the length of follow-up in previous studies ranged from 30 days to 12 months after TAVR. Research studies that report long-term outcomes beyond 1 year are scarce.

Therefore, the aim of our study was to compare the mortality rate, event rate, and functional capacity in AS patients with BcAV after SAVR and TAVR, and analyze the longer term outcomes.

2 | METHODS

2.1 | Study design and population

We performed a single-center, cross-sectional study on patients with a BcAV who underwent SAVR or TAVR from July 2013 to August 2018 at the Cheng Hsin General Hospital. The patients were matched 1:2 for age and sex. This study was approved by the Institutional Review Board (IRB) and each patient signed an informed consent form before the procedure.

2.2 | Baseline risk score

Baseline characteristics were collected to investigate the influence of sociodemographic and medical characteristics, including age, sex, diagnosis, weight, and height. Furthermore, the STS risk score, which is useful for prediction of the risk of mortality and various morbidities following surgery, was calculated using the online STS score calculator.[9]

2.3 | Endpoints

The main endpoint was a major adverse cardiovascular event (MACE) following SAVR and TAVR. Events recorded included non-cardiovascular (CV) mortality, CV mortality, recurrent nonfatal disabling stroke, recurrent nonfatal myocardial infarction (MI), and important adverse events. The important adverse events included heart failure, recurrence of angina pectoris, and repeat percutaneous coronary intervention or coronary artery bypass graft.[10] The secondary outcome was functional recovery 6 months after aortic valve surgery, which was defined as the metabolic equivalent (MET) determined according to the maximum activity level that was ascertained during a phone follow-up.

2.4 | Statistical analyses

Independent *t* tests and χ^2 tests were used to compare basic characteristics between the SAVR and TAVR groups. The independent predictors of mortality, MACE, and functional recovery were determined using a logistic regression model adjusted for STS score, left and right ventricular ejection fraction, diabetes, hypertension, and age. A Kaplan–Meier analysis and log-rank test were used to analyze the survival rate and the rate of stroke and MI in SAVR and TAVR patients. Probability values of less than .05 were considered significant. All analyses were performed using SPSS (version 19.0, SPSS Inc.).

3 | RESULTS

3.1 | Patient characteristics

Eighty-two patients who underwent SAVR and 48 patients who underwent TAVR were included in the study. All SAVR procedures were done through a full sternotomy, and TAVR procedures were done through a transfemoral approach. All patients received general anesthesia and intubation during the procedure, and a transesophageal echocardiogram was routinely used to assess the final result for both groups. Of these, the mean age was 56.8 ± 26.9 years, 35% were women, and the mean STS score was 3.29 ± 4.69. Before the procedure, most patients were classified as New York Heart Association (NYHA) functional classes II and III (49% and 33% of patients, respectively). The demographic and clinical characteristics of the two groups are shown in Table 1.

3.2 | Follow-up on mortality and rate of events

The mean follow-up time for the 130 patients was 2.86 ± 1.47 years, during which time there were nine deaths (7%). One patient suffered a recurrent nonfatal stroke (0.8%), and four patients suffered nonfatal MI (3.1%). The mean intensive care unit stay was 2.3 ± 1.7 days,

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TABLE 1 Baseline characteristics of the study population

	All (n = 130)	TAVR (n = 48)	AVR (n = 82)
Demographic			
Age (years)	56.7 ± 26.9	65.7 ± 7.8	51.5 ± 32.3
Gender (male)	85 (65.4%)	28 (58.3%)	57 (69.5%)
BMI (kg/m^2)	24.9 ± 4.2	24.1 ± 3.7	25.4 ± 4.4
STS PROM	3.3 ± 4.7%	6.25 ± 6.4	1.56 ± 1.7*
Pre-op NYHA class			*
III	40 (30.8%)	22 (45.8%)	18 (22.0%)
IV	9 (6.9%)	2 (4.2%)	7 (8.5%)
Echocardiography			
PPG (mmHg)	74 ± 39	84 ± 38	67 ± 39*
MPG (mmHg)	46 ± 28	51 ± 25	42 ± 29
AVA (cm^2)	0.7 ± 0.4	0.7 ± 0.3	0.7 ± 0.4
LVEF	48 ± 12	46 ± 14	50 ± 10*
RVEF	52 ± 8	51 ± 9	53 ± 7
RVSP (mmHg)	36 ± 14	41 ± 16	34 ± 11*
Past medical history			
HTN	69 (53.1%)	29 (60.4%)	40 (48.8%)
DM	26 (20.0%)	16 (33.3%)	10 (12.2%)*
Heart failure	33 (37.9%)	15 (31.3%)	18 (22.0%)
MI	4 (3.1%)	2 (4.2%)	2 (2.4%)
PCI	17 (13.1%)	14 (29.2%)	3 (3.7%)
CABG	1 (1.1%)	1 (2.1%)	0 (0.0%)
PPM	5 (5.7%)	5 (10.4%)	0 (0.0%)
AF	15 (17.2%)	8 (16.7%)	7 (8.5%)
Stroke	8 (9.2%)	7 (14.6%)	1 (1.2%)

Note: Data are presented as mean ± *SD* or absolute values (%).

Abbreviations: AF, atrial fibrillation; AVA, aortic valve area; BMI, body mass index; CABG, coronary artery bypass graft; DM, diabetes mellitus; EF, ejection fraction; HTN, hypertension; LV, left ventricular; LVEF, left ventricular ejection fraction; MI, myocardial infarction; MPG, mean pressure gradient; NYHA, New York Heart Association Functional Classification; PCI, percutaneous coronary; PPG, peak pressure gradient; PPM, permanent pacemaker; Pre-op, before operation; PROM, predicted risk of mortality; RV, right ventricular; RVSP, right ventricular systolic pressure intervention; STS, society of thoracic surgeons.

*$p < .05$, groups comparison.

and the mean total hospitalization was 17.3 ± 11.1 days. After adjustment for modifiers, the aortic procedure was not associated with mortality or MACE during the follow-up period. The procedural and clinical outcomes are indicated in Table 2.

3.3 | Correlation between procedure type and functional recovery

Logistic regression analyses indicated that SAVR was a significant predictor of functional recovery. Patients who underwent SAVR showed a higher rate (87.8%, p = .000) of functional recovery (>3 METs) and showed a significantly higher odds ratio (3.56; 95% CI, 1.19–10.63, p = .02) for the increased recovery of functional capacity compared with patients who underwent TAVR (Table 3). The Kaplan–Meier survival curves in Figure 1 show no difference in survival rates based on the aortic procedure.

4 | DISCUSSION

Nowadays, controversy remains as to whether SAVR is the gold standard for AS patients with BcAV. Some recent studies have suggested that TAVR may be a safe treatment modality in selected AS patients with BcAV because of technical enhancements and new-generation devices that are associated with lower failure rates and less postprocedural complications.[7,8] In our study, we divided our patients into two groups according to the surgical procedure type. Contrary to previous arguments, our results support the idea that, compared with SAVR, TAVR is a reasonable option for AS patients

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TABLE 2 Procedural and clinical outcomes

	All (n = 130)	TAVR (n = 48)	AVR (n = 82)
ICU stay, days	2.3 ± 1.7	2.2 ± 2.4	2.4 ± 1.0
Hospital stay, days	17.3 ± 11.1	17.3 ± 14.9	17.3 ± 8.2
MACE			
Nonfatal stroke	1 (0.8%)	1 (2%)	0 (0.0%)
Nonfatal MI	4 (3.1%)	0 (0.0%)	4 (4.9%)
Non-CV mortality	9 (6.9%)	5 (10.4%)	4 (4.9%)
CV mortality	1 (0.8%)	1 (2.1%)	0 (0.0%)
Bleeding, major	0 (0.0%)	0 (0.0%)	0 (0.0%)
PVL, moderate or worse	6 (4.6%)	6 (12.5%)	0 (0.0%)
Functional recovery			
Walk dependently	18 (13.8%)	15 (31.3%)	3 (3.7%)*
Wheelchair reliance	13 (10.0%)	11 (22.9%)	2 (2.4%)*
Bed ridden	5 (3.8%)	3 (6.3%)	2 (2.4%)

Note: Data are presented as mean ± *SD* or absolute values (%).

Abbreviations: CV, cardiovascular; ICU, intensive care unit; MACE, major adverse cardiac events; MI, myocardial infarction; PVL, paravalvular leak.

*p < .05, groups comparison.

TABLE 3 Logistic analysis of factors that might influence functional recovery in AS patients with BcAV after aortic procedures

	B	OR	95% CI	*p*
Surgical procedure type, SAVR*	1.27	3.56	1.19–10.63	**.02**
STS PROM	−0.01	0.99	0.80–1.23	.93
Pre-op NYHA class	−0.18	0.84	0.19–3.78	.82
Age	−0.16	0.85	0.72–3.78	.85
LVEF	0.11	1.12	0.97–1.28	.11
RVSP	−0.01	0.99	0.95–1.04	.73
DM	−0.12	0.89	0.28–2.82	.23

Note: Data presented with *p* values and odds ratio (OR) with 95% CI. Bold value statistically significant at p < .05.

Abbreviations: DM, diabetes mellitus; LVEF, left ventricular ejection fraction; NYHA, New York heart Association Functional Classification; PROM, predicted risk of mortality; Pre-op, before operation; RV, right ventricular; RVSP, right ventricular systolic pressure; SAVR, surgical aortic valve replacement; STS, society of thoracic surgeons.

*p < .05.

with BcAV with intermediate surgical risk. Moreover, SAVR may not be the gold standard strategy for this special group because our main finding showed that there was no significant difference in mortality rates and event rates in AS patients with BcAV in the SAVR and TAVR groups over a mean follow-up period of 2.86 ± 1.47 years. This was despite the higher STS scores, which indicated a bigger surgical risk, and the lower left ventricular ejection fraction (LVEF) measurements, which represented poorer heart condition, in the TAVR group.

This study is the first analysis reporting the results of functional capacity in AS patients with BcAV who underwent SAVR and those who underwent TAVR. Six months after surgery, TAVR was associated with a significantly lower functional capacity, and we concluded that SAVR is the main predictor of better functional recovery. These findings indicate that patients who received TAVR had higher STS scores, indicating that they had a higher risk of comorbidity before TAVR, resulting in poorer activity levels.[11] This may be because the lower LVEF contributes to weak muscle strength and decreased activity level as a result of insufficient perfusion from the low cardiac output. In addition, we also found that after 6 months, AS patients with BcAV were not able to walk independently and had a higher reliance on a wheelchair. Therefore, the heart team needs to focus on functional recovery and quality of activity when TAVR is chosen for AS patients with BcAV.

Jette et al.[12] reported that the MET concept provides a feasible method of describing exercise tolerance and functional capacity. Activities of daily living that demanded approximately three METs were defined as light intensity and included bed mobility, casual walking, and light housework such as dishwashing, ironing, and cooking.[12] Thus, patients showed higher functional recovery after SAVR, especially for activities demanding over three METs, needed less assistance, and were much independent in their basic daily life. Moreover, higher functional capacity is associated with increased exercise tolerance and decreased risk of future cardiovascular disease and events, according to a scientific statement from the American Heart Association.[13] Accordingly, for AS patients with BcAV, for the capability in daily living and better prognosis after TAVR, strategies to improve functional capacity, such as cardiac rehabilitation, should be arranged without delay and should be prescribed preoperatively.

5 | STUDY LIMITATIONS

There were two limitations associated with this study that warrant mention. First, the sample size of our study was relatively small even though we included all AS patients with BcAV who underwent TAVR from 2013 to 2018. Increasing the number of patients was difficult because TAVR is still not a gold standard treatment for these special groups. Second, our patients were younger males with an NYHA level II to III and an intermediate risk of surgery. Therefore, the results of our analysis may not apply to other populations.

6 | CONCLUSIONS

For AS patients with BcAV, there were no significant differences in the rate of MACE in those who underwent SAVR and those who underwent TAVR during the 3-year follow-up period. Our analysis showed that SAVR was a significant predictor of functional recovery and patients who underwent TAVR showed a lower level of

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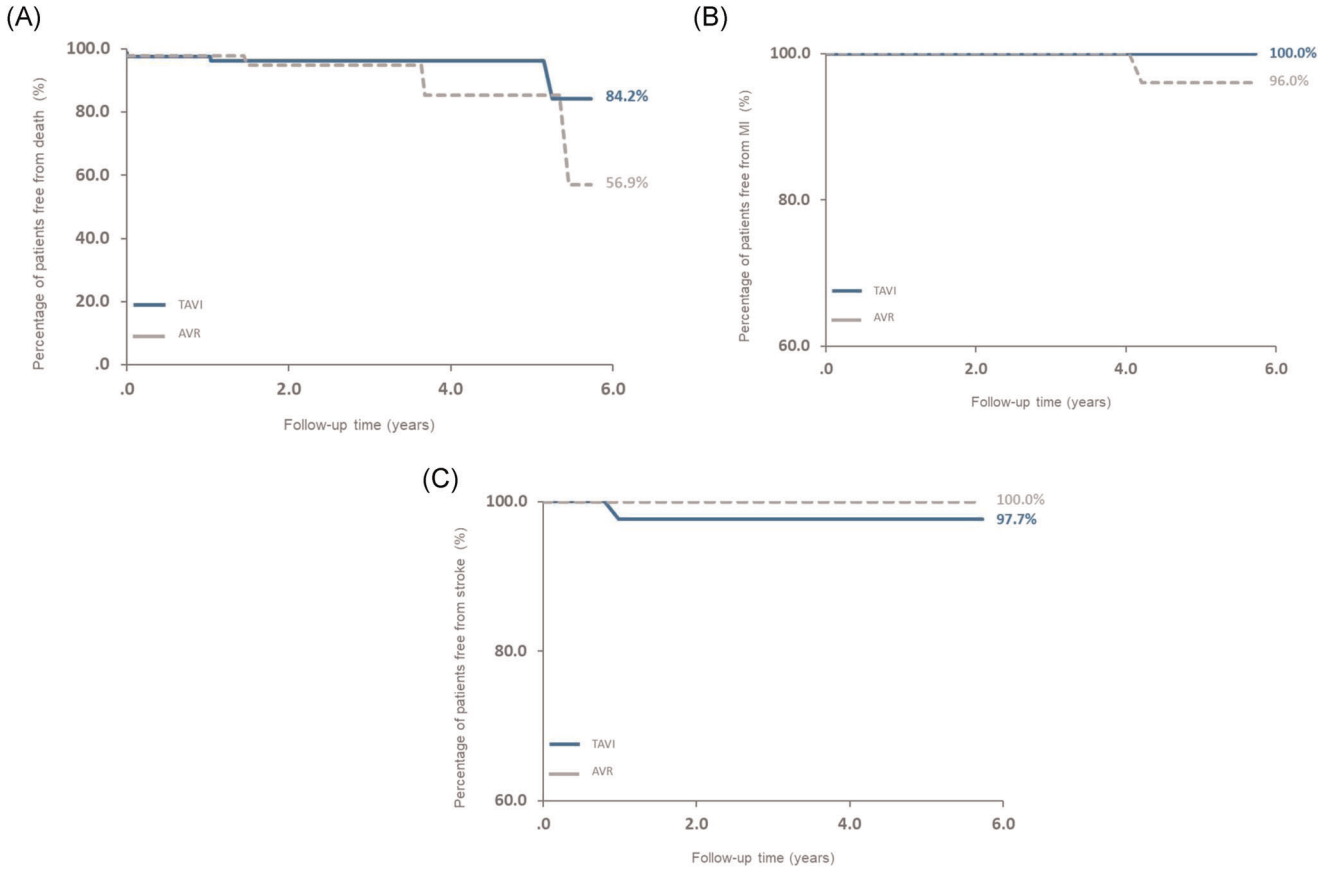

FIGURE 1 The Kaplan–Meier survival curves show no significant difference in the rates of survival, MI, and stroke based on TAVR and SAVR. MI, myocardial infarction; SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement

functional capacity, especially 6 months after surgery. In summary, TAVR is an acceptable option for AS patients with BcAV with intermediate surgical risk. Furthermore, for better independence in daily living and better prognosis, early intervention aimed at improving functional capacity is highly recommended and future studies are required to determine the most effective programs for this special group of patients.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author, Huang, Hsin-Yi, upon reasonable request.

ORCID

Hui-Yu Tsai http://orcid.org/0000-0003-2090-6005
Yu-Shan Lin http://orcid.org/0000-0003-4816-9887

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How to cite this article: Tsai H-Y, Lin Y-S, Wu I-C, et al. Major adverse cardiac events and functional capacity in patients at intermediate risk undergoing transcatheter versus surgical aortic valve replacement for aortic stenosis with bicuspid valves. *J Card Surg*. 2021;36:828-833.
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